

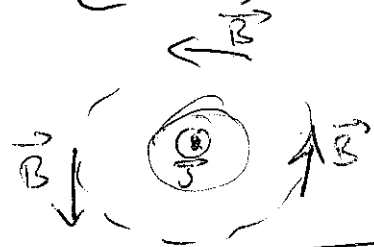
* FARADAY'S LAW IN INTEGRAL FORM

$$\oint_C \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int_{\Sigma} \vec{B} \cdot d\vec{S} = - \int_{\Sigma} \left(\frac{\partial \vec{B}}{\partial t} \right) \cdot d\vec{S}$$

Where the curve C encloses the surface Σ .

$\Rightarrow$ ~~Comparison with~~ Ampere's law for magnetostatic fields?

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc.}} = \int_{\Sigma} (\mu_0 \vec{J}) \cdot d\vec{S}$$



~~enclosed~~ current causes "curly" magn-field

$$\oint_C \vec{E} \cdot d\vec{l} = - \int_{\Sigma} \left(\frac{\partial \vec{B}}{\partial t} \right) \cdot d\vec{S}$$



enclosed ~~time-dependent~~ time-dependent magnetic field causes "curly" electric field.

* TWO TYPES OF ELECTRIC FIELDS

$$\text{EMF } \mathcal{E} = \oint_C \vec{E} \cdot d\vec{l}$$

Q. Why didn't we just say voltage = "potential difference"

5C

A: Because potential is not defined.

$\Rightarrow$ Definition $\vec{E} = -\vec{\nabla} V$ requires ~~$\vec{\nabla} \times \vec{E} = 0$~~

Now we have $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$; ~~$\vec{\nabla} \times \vec{E} = 0$~~

(studying physics beyond electrostatics.)

In electrostatics, $\oint \vec{E} \cdot d\vec{l} = 0$, because



$$\oint \vec{E} \cdot d\vec{l} = V_A - V_A = 0$$

(potential difference of a point with itself.)

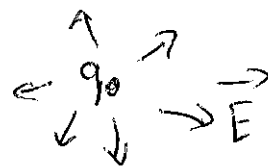
2 TYPES of $\vec{E}$ fields

Electric

Fields created in electrostatics:

$\rightarrow$ field lines end or start at charges.

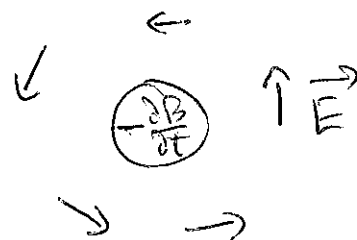
$\rightarrow$ Zero curl.



Electric fields created by induction (changing $\vec{B}$):

$\rightarrow$ field lines loop around onto themselves

$\rightarrow$ no divergence.



* Maxwell's 4 equations

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \text{electromagnetic induction}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

* We derived the term

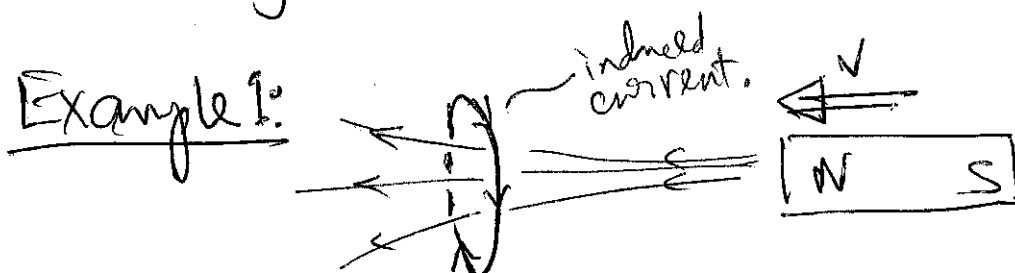
$$\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$$

by imposing the continuity equation. This quantity is called the **DISPLACEMENT CURRENT**

* LENZ'S LAW (fixes the direction of induced ~~EMF~~/current)

~~currents~~ currents induced by ~~change~~ change in magnetic flux ~~oppose the change, through the B-field created by the induced current.~~

oppose the change, through the $\vec{B}$ -field created by the induced current.



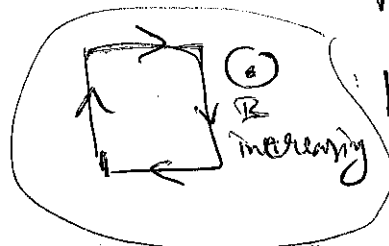
Flux increasing. Induced current will **OPPOSE** increase.
 $\Rightarrow$ Will point rightward.

Example 2a

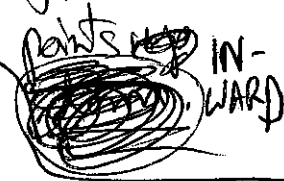
$$|\vec{B}| = \alpha t$$

increasing $|\vec{B}|$

induced current opposes increase



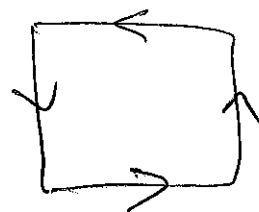
~~induced field points down~~
induced field points up

Example 2b

decreasing $|\vec{B}|$, e.g. $|\vec{B}| = B_0 e^{-\alpha t}$

induced current opposes decrease

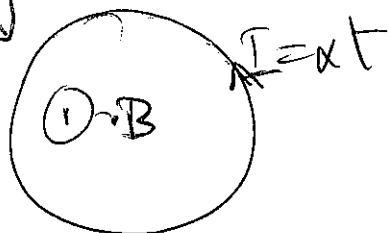
$\Rightarrow$ ~~induced field points down~~ field due to induced current points UP.

Example 3

(Without ~~a circuit/wire~~ a circuit/wire!)

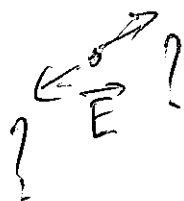
Changing current in Solenoid

CROSS-SECTION OF SOLENOID



Increasing solenoid current
 $\Rightarrow$ increasing $\vec{B}$ -field / flux

$$B(t) = \mu_0 n I(t) = \mu_0 n \alpha t$$



Which direction is the induced $\vec{E}$ -field?

$$\oint \vec{E} \cdot d\vec{l} = \int \left(-\frac{\partial B}{\partial t} \right) \cdot d\vec{A}$$

Answer 1 ~~the~~ imagine a wire loop

circling around solenoid. Induced EMF should
OPPOSE INCREASE of flux $\Rightarrow$ ~~the~~ imagined
induced ~~current~~ ^{current} should create $\vec{B}$ -field downward.

$\Rightarrow$ ~~the~~ induced $\vec{E}$ -field points CLOCKWISE,
opposite direction of increasing current in solenoid.

* Lenz's law corresponds to ^{negative} Δ SIGN in ~~the~~ Faraday's
law.

$\rightarrow$ Lenz's law makes sense energetically,
because ~~other~~ ^{other} sign would lead to perpetual
energy production.

* EXERCISE:



~~the~~ Loop is being deformed
so that loop area DECREASES.
 $\vec{B}$ is constant and $\perp$ to plane.

Find the ~~area~~ ^{direction} of the induced current.